

Matching User Preferences: Optimal Assignment of Students to Projects

Project Plan

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Note on the planning. This timeline is a preliminary, approximate version of the project plan. The spring break fell right after our first meeting with our supervisor, which limited the amount of scoping we could do collectively before the Project Plan deadline. The plan presented here may therefore be revised and complemented once we resume work after the break, in particular regarding the precise scope of the Phase 2 extensions and the allocation of tasks among group members.

1 Project Description

The matching problem we tackle is the assignment of students to projects, where each student submits an ordinal preference list over the available projects and the school must produce a final assignment that is satisfactory and fair. For the latest DSAI project assignment, this allocation was solved using a Linear Program. Now, our goal is to implement and compare classical algorithms, evaluate them on the same dataset and explore extensions that go beyond the standard one-shot, one-sided formulation.

Our main theoretical groundwork is Chapter 14 of the *Handbook of Computational Social Choice* (Klaus, Manlove, Rossi), which classifies bipartite matching problems with preferences into one-sided (House Allocation, HA) and two-sided (Hospitals/Residents, HR) variants and presents the canonical algorithms.

1.1 Tasks

Phase 1: Classical mechanisms (weeks 1–5, up to the midterm). Read Chapter 14 and implement the three canonical algorithms covered there: Serial Dictatorship (**SD**), Gale-Shapley deferred acceptance (**GS**, both applicant and proposer-oriented variants), and Top Trading Cycles (**TTC**). Define the evaluation metrics (stability function and cost function): a *cost function* aggregating individual ranks (e.g. sum of obtained ranks, number of unmatched students, Borda welfare), and a *stability score* counting blocking pairs when one side has priorities (relevant when the school ranks students by GPA). Run all three mechanisms on the course data and on synthetic instances. The midterm evaluation on May 29 will be the deadline for this phase.

Phase 2: Extensions and analysis (weeks 6–8). Once the baselines are in place, we explore variants of the problem and propose our own approaches. Candidate directions, to be narrowed down at the end of Phase 1:

- **Two-sided preferences with priorities:** the school also ranks students (e.g. by GPA), turning the problem into an instance of HR; compare the student-optimal and school-optimal stable matchings.
- **Partial preferences and ties:** truncated preference lists and indifferences (HRT-style instances), with appropriate tie-breaking rules.
- **Veto**es and **constraints:** forbidden student–project pairs, project quotas, minimum group sizes.
- **Coalitions and groups:** students who must be assigned together, motivated by team projects.
- **Statistical setting:** treating reported preferences as noisy samples from an underlying probability distribution

Phase 3: Synthesis (week 9). Final report and oral presentation by the June 26 deadline: consolidate results, articulate trade-offs, and give a recommendation for which mechanism (or combination) we believe best fits the Télécom Paris use case.

1.2 Data

Two data sources will be used in parallel:

- **Real data** collected within the latest DSAI Project matching: anonymised preference lists submitted by students over the available projects.
- **Synthetic data** generated under controlled preference models

2 Deliverables

- Git repository with the implementations of SD, GS, TTC, and the Phase 2 extensions.
- Reproducible experimental notebooks comparing the mechanisms on the course data and on synthetic benchmarks.
- Final report describing the methods and results
- Oral presentation of the project.